

Ultimate Nutrition

TestostroGROW 2 HP

Testosterone is much more than a sex hormone. Testosterone improves oxygen uptake throughout the body, helps control blood sugar, regulate cholesterol and maintain immune surveillance. Mitochondria use testosterone for Krebs's cycle to occur and energy to be produced. There are testosterone receptor sites in cells throughout the body, most notably in the brain and heart. Protein synthesis for maintaining muscle mass and bone formation requires testosterone. Testosterone decreases body fat and enhances lean muscle mass.

Testosterone enhances aerobic metabolism. This has the result of improving muscle tone, burning fat, and improving energy. Indirectly, it is suggested this increase in testosterone also effects mood. Studies have clearly shown that increased testosterone levels have a beneficial effect on overall health, weight, libido, and energy in men. Each ingredient in TestostroGROW 2 HP works toward the same goal: to potentiate testosterone. Each ingredient follows a different path to achieve this result.

The bioflavonoid chrysin has shown potential as a natural aromatase-inhibitor. Studies demonstrate that chrysin is an extremely potent inhibitor of aromatase activity. Chrysin, is also a potent antioxidant that possesses vitamin-like effects in the body. It has been shown to induce an anti-inflammatory effect, possibly through inhibition of the enzymes 5-lipoxygenase and cyclooxygenase inflammation pathways. Chrysin may offer libido-enhancing effects as well by increasing free testosterone, decreasing excess estrogen and producing an anti-anxiety effect.

Tribulus terrestris is a plant that grows in many tropical and moderate areas of the world. It is the steroidal saponins, particularly the dominant saponin protodioscin, that are responsible for the pharmacological activities of tribulus. Tribulus terrestris boosts testosterone levels by stimulating the increase of luteinizing hormone (LH). When LH levels are increased, the natural production of testosterone also increases.

Cucurbita Pepo and Saw Palmetto Extract are another resource to prevent testosterone from becoming "deactivated." The Cucurbita Pepo seeds' principal constituent inhibits the conversion of testosterone into dihydrotestosterone. As a natural source of zinc, Cucurbita Pepo also helps to optimize the production of testosterone. Saw palmetto berries consist of of a volatile oil, which is high in phytosterols. These ingredients not only control the conversion of testosterone to DHT (DHT is formed when the male hormone testosterone interacts with the enzyme 5-alpha reductase) but also block it's binding to the prostate cells, which reduce the levels of inflammation in the prostate. Saw palmetto also has anti-estrogenic and anti-androgenic effects.

When testosterone binds to SHBG (sex hormone-binding globulin), it loses its biological activity and becomes known as "bound testosterone," as opposed to the desirable "free testosterone." Nettle root extract provides a unique mechanism for increasing levels of free testosterone. Constituents of nettle root bind to SHBG in place of testosterone, thus reducing SHBG's binding of free testosterone. Nettle also inhibits the binding of DHT to attachment sites on the prostate membrane and inhibits enzymes such as 5 alpha reductase that cause testosterone to convert to DHT. The combination of Nettle Root with Avena Sativa also leads to an increase of aerobic power and muscle strength.

Avena sativa is a botanical extract that has traditionally been used to increase strength, mind, spirit and body. It has also been found to free up testosterone that's stuck to other compounds in your body. Avena sativa is a recognized mood enhancer and has traditionally been used as a remedy to help with recovery from exhaustion and low mood. It works well as an addition to tribulus, enhancing feelings of well being and restoring energy.

Researchers have observed that metabolism of estrogen occurs via one of two pathways: The 'harmful' metabolic pathway, 16 alpha-hydroxylation, or the 'beneficial' metabolic pathway, 2-hydroxylation. Indole-3-Carbinol (I3C) can shift the production of estrogen metabolites away from dangerous 16-hydroxy in favor of beneficial 2-hydroxy metabolites. Indole-3-carbinol is converted by stomach acid to diindolymethane (DIM) and indole (3,2,b) carbazole (ICZ). DIM produces more 'good' estrogen metabolites, which compete with testosterone for protein binding, helping to maintain a higher free testosterone level. The 'good' estrogen metabolites along with increased free testosterone promoted by DIM increase fat mobilization and a fat-burning metabolism.

Another reason testosterone production can decline is because of oxidative damage directed at the tissues that synthesize testosterone. Antioxidants support testosterone production. This is where Daidzein and Alpha Lipoic Acid do their best work. Daidzein is an isoflavone. This compound, found only in plants, has antioxidant activity. Alpha Lipoic Acid also acts as an antioxidant, repairing a wide array of free radical damage at the cellular level, keeping cells healthy and aiding in nutrient absorption. Alpha Lipoic Acid aids in the conversion of carbohydrates into energy, improves metabolism and normalizes blood sugar.

Hawthorne Berry is rich in flavonoids. It has the ability to dilate coronary blood vessels (the vessels supplying the heart with vital oxygen, blood, and nutrients), and strengthen the heart's pumping ability. Some of the flavonoids increase the force of contraction of the heart by inhibiting the enzyme cAMP phosphodiesterase (the enzyme responsible for breaking down cyclic adenosine monophosphate). Cyclic AMP is an intracellular secondary messenger of luteinizing hormone (LH).

TestostroGrow 2 HP is a versatile time-release anabolic stack that combines the most effective testosterone stimulators available. It has been precisely formulated to potentiate testosterone in order to ensure a steady, nonstop state of increased testosterone for more successful anabolic results.

SELECTED REFERENCES:

- Campbell DR, Kurzer MS. J Steroid Biochem Mol Biol. Flavonoid inhibition of aromatase enzyme activity in human preadipocytes. 1993 Sep;46(3):381-8.
- Edmunds KM, Holloway AC, Crankshaw DJ, Agarwal SK, Foster WG. The effects of dietary phytoestrogens on aromatase activity in human endometrial stromal cells. Reprod Nutr Dev. 2005 Nov-Dec;45(6):709-20.
- He F, Feng L. Effects of some micronutrients on partial androgen deficiency in the aging male. Zhonghua Nan Ke Xue. 2005 Oct;11(10):784-6.
- Jeong HJ, Shin YG, Kim IH, Pezzuto JM. Inhibition of aromatase activity by flavonoids. Arch Pharm Res. 1999 Jun;22(3):309-12.
- Kao YC, Zhou C, Sherman M, Laughton CA, Chen S. Molecular basis of the inhibition of human aromatase (estrogen synthetase) by flavone and isoflavone phytoestrogens: A site-directed mutagenesis study. Environ Health Perspect. 1998 Feb;106(2):85-92.
- Kellis JT Jr, Vickery LE. Inhibition of human estrogen synthetase (aromatase) by flavones. Science. 1984 Sep 7;225(4666):1032-4.
- Monteiro R, Azevedo I, Calhau C. Modulation of aromatase activity by diet polyphenolic compounds. J Agric Food Chem. 2006 May 17;54(10):3535-40.
- Packer L, Witt EH, Tritschler HJ. alpha-Lipoic acid as a biological antioxidant. Free Radic Biol Med. 1995 Aug;19(2):227-50.
- Ruiz-Larrea MB, Mohan AR, Paganga G, Miller NJ, Bolwell GP, Rice-Evans CA. Antioxidant activity of phytoestrogenic isoflavones. Free Radic Res. 1997 Jan;26(1):63-70.

FREQUENTLY ASKED QUESTIONS:

1: What is I3C?

Indole-3-Carbinol (I3C) is a naturally occurring phytochemical found in cruciferous vegetables such as cabbage, broccoli, and kale. I3C provides powerful antioxidant protection, helping to prevent cellular damage caused by free radicals. In addition, I3C is converted to DIM in the body and can help to maintain healthy normal hormonal balance.

2: Where is Hawthorne Berry found and what is it used for?

Hawthorne Berry (crataegus oxyacantha) comes from the hawthorne shrub, a spiny tree that is indigenous to the Mediterranean region. Hawthorne Berry has a primary chemical

constituent that includes vitamin C, flavonoids (quercetin, quercetrin), glycosides, proanthocyanidins, anthocyanidins, saponins, tannins, and cratetegin (most prevalent in the flowers, then leaves, then berries). Since the 19th century hawthorne berries have been used to support the heart, and to normalize cardiovascular functions. Hawthorne berry helps protect the arterial walls, can dilate (enlarge) coronary blood vessels (the vessels supplying the heart with vital oxygen, blood, and nutrients), and strengthens the heart's pumping ability. They inhibit cAMP phosphodiesterase; the enzyme responsible for breaking down cyclic adenosine monophosphate (a secondary messenger of luteinizing hormone).